

FOO TECHNICAL RESPONSE

Bede – Loan Management System

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Version 1.0.0



B2B Fintech Technology and Services Provider enabling Banks, FinTechs, Retailers and Service Providers

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Company Profile

Our Mission is **INNOVATION. EXPERTISE. QUALITY.**

Founded in 2009, **FOO** has emerged as a leading Fintech provider by focusing on Technology, Innovation, Expertise and Quality.

Working with leading clients in their industries, **FOO** has quickly expanded in the MENA region and has set the benchmark for design experience and quality.

FOO's vision is to expand globally and create a positive impact on future technologies.

Check **Annex 1** for full **Company Profile**



Introduction

Executive Summary

BEDE, referred to as Client, has shown interest in an LMS solution to provide the clients with a solution for loan disbursement.

In this regard, FOO will offer its **FOO LMS** which incorporate cutting-edge fintech features and can be seamlessly customized to align with the client's design requirements.

FOO's solutions have revolutionized the fintech industry, transforming the way people interact with financial institutions and conduct their day-to-day financial activities.

In this proposal we will be discussing the below main features offered through the mobile app and different portals as per the below:

FOO Products

- FOO Loan Management System



Client Requirements

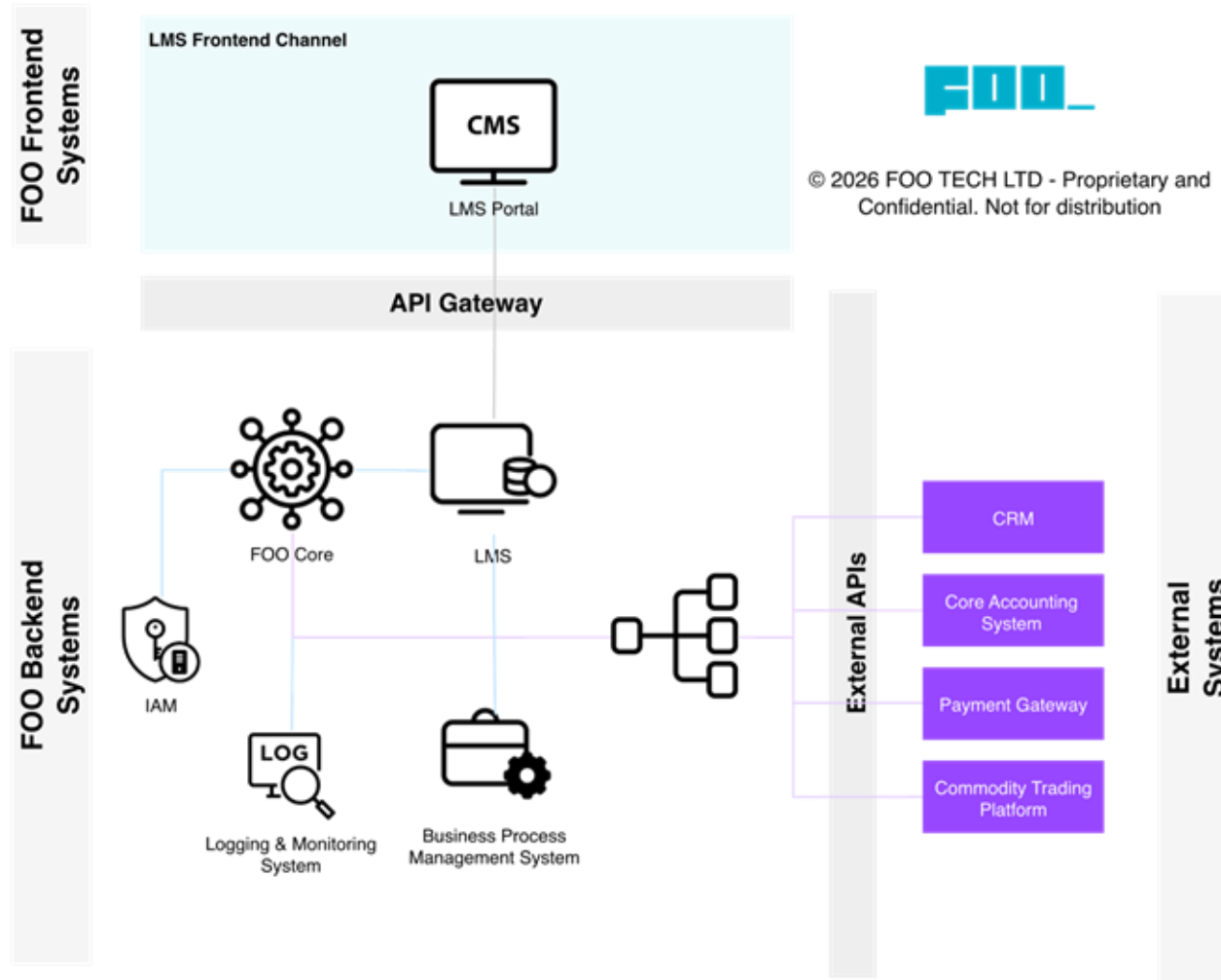
The following section details all requirements that have been provided by the Client pertaining to the current solution.

FOO technical proposal and the solution provided will be based on fulfilling the requirements shared in the RFP documents.



Proposed Solution

Solution Architecture



FOO LMS Portal

Functionality	Details	Implementation Type
Loan Origination	Loan product configuration with customizable parameters including loan amount, tenure, and interest rate.	Out of the box
	Document management for storing and verifying borrower information	Out of the box
Loan Disbursement and Repayment	Manual and scheduled loan disbursement processing.	Out of the box
	Flexible repayment schedules and options.	Out of the box
Profit Calculation Methods	Support for flat profit calculation methodology	Out of the box
	Real-time profit calculation and tracking for accurate reporting	Out of the box
Grace Period	The system allows for the inclusion of grace periods, during which clients are not required to make repayments, providing them with flexibility in managing their loans.	Out of the box
	Configuration of grace periods including principal-only grace, Interest-only grace and full grace periods	Out of the box
Penalty and Fees Configuration	Configuration of fees and penalties including fixed and percentage-based charges.	Out of the box
	Support for late payment penalties, upfront fees, and recurring charges.	Out of the box
	Automatic application of penalties based on overdue installments.	Out of the box
Loan Approval Workflow	Configurable loan approval process including manual approval and maker-checker workflows. Maker-Checker flows applies on all features that have a permission linked to it in the Permissions/Roles section	Enhanced by FOO
	Loan approval based on predefined product rules and limits	Out of the box
Portfolio Management	Real-time tracking of loan portfolio including outstanding balances, repayment status, and arrears.	Out of the box
	Loan lifecycle management including active, closed, and written-off statuses.	Out of the box
	Basic delinquency tracking using Days Past Due (DPD).	Out of the box
Loan Account Management	Detailed loan account view including repayment schedule, transaction history, charges, and balances.	Out of the box
	Support for multiple loans per customer.	Out of the box
	Full transaction audit trail at loan level	Enhanced by FOO
Repayment	Processing of scheduled and ad-hoc repayments with automatic allocation to penalties, fees, interest, and principal.	Out of the box
	Support for overpayments and prepayments	Out of the box

FOO LMS Portal

Functionality	Details	Implementation Type
Delinquency Management	Tracking of overdue loans and arrears.	Out of the box
	Calculation of Days Past Due (DPD) and basic aging buckets.	Out of the box
	Support for basic delinquency bucket classification based on DPD ranges (e.g., 1–30 days, 31–60 days, 61–90 days, 90+ days)	Out of the box
Loan Rescheduling & Modifications	Support for loan rescheduling including change of repayment dates and extension of loan term.	Out of the box
Write-Off Management	Support for full and partial loan write-offs.	Out of the box
	Automatic adjustment of loan balances upon write-off.	Out of the box
	Recording of write-off transactions with audit trail.	Enhanced by FOO
Early Settlement / Prepayment	Support for early loan settlement through full repayment of all outstanding amounts (principal, interest, fees, and penalties).	Out of the box
	Loan closure is system-driven and occurs automatically once all outstanding balances are cleared.	Out of the box
Partial Settlement	Ability to pay one or multiple future installments in advance.	Out of the box
	Support for partial installment payments.	Out of the box
Accounting Entries	Automatic generation of accounting entries for disbursements, repayments, fees, and write-offs.	Out of the box
	Integration with accounting systems via APIs.	Enhanced by FOO
Reporting	Standard reports including loan portfolio, repayment schedules, arrears, and transaction reports.	Customized by FOO
	Export capabilities (CSV/Excel).	Customized by FOO
Audit Trail	Tracking of all loan transactions and user actions at transaction level.	Enhanced by FOO
	Visibility of historical changes and operations.	Enhanced by FOO
VAT/TAX Configuration	Ability to define tax components as part of charges (fixed or percentage-based).	Enhanced by FOO
	Automatic calculation and application of VAT during fee/penalty posting.	Enhanced by FOO
APIs		

FOO LMS Portal

Functionality	Details	Implementation Type
Advanced Loan Product Configuration	Support for multiple amortization methods (equal installments, equal principal, etc.)	Out of the box
	Support for multi-disbursement loans (tranches)	Out of the box
	Configurable repayment frequency (daily, weekly, monthly, etc.)	Out of the box
	Holiday calendar setup and configuration affecting repayment schedule	Out of the box
Custom Fields & Metadata Configuration	Ability to define and attach additional custom fields to core entities such as Clients and Loans.	Enhanced by FOO
	Supports extension of standard data model without code changes. Enables capturing institution-specific information using configurable data tables.	Enhanced by FOO
Insurance Payment Handling	Support for insurance payment collection as part of financing	Customized by FOO
	Ability to configure insurance amount as a fee linked to the financing account	Customized by FOO
	Insurance fee can be included within installments or applied separately	Customized by FOO
Commodity Transaction Integration	Ability to select and link a commodity transaction during financing creation	Customized by FOO
	Retrieve list of available commodity transactions from external system	Customized by FOO
	Store commodity reference and key details with the financing account	Customized by FOO
	Enable tracking of commodity-linked financing for reporting and audit	Customized by FOO
Automatic Payment via Standing Instructions (SI)	Support automatic payment of financing installments using Standing Instructions	Enhanced by FOO
	Ability to configure SI for fixed or due-based payments	Enhanced by FOO
	Trigger automatic transfer from linked account on scheduled dates	Enhanced by FOO
	Enable tracking of SI execution and payment status	Enhanced by FOO
Murabaha Financing Type	Introduced new "Financing Type" concept to support Islamic financing	Enhanced by FOO
	Profit Handling (Interest → Profit Relabeling)	Customized by FOO
	Introduced new financial fields and validation rules for Murabaha (Cost, Profit, Selling Price...)	Customized by FOO

Deliverables

Deliverable Area	Description Summary
Project Kickoff & Governance Setup	Initiation of the engagement including the kickoff workshop, project charter, governance model, RACI, communication plan, and risk/issue logs. Establishes the PMO tooling (Jira, Slack, shared drive) and confirms stakeholder roles on both sides before any discovery or design work begins.
Business Requirements Document (BRD) - Capped	Structured requirements gathering through workshops with BEDE business teams to document the full post-disbursement lifecycle, product configuration parameters, and business rules for aging, penalties, settlement, and restructure. Culminates in a signed BRD that locks the functional scope before SRS work begins.
System Requirements Specification (SRS) & Test Cases	Translation of the signed BRD into a technical specification covering screen-level specs for custom LMS portal changes, and QA test cases reviewed with BEDE. Acts as the formal build contract and is signed off before implementation starts.
Dev / UAT / Prod Environment Setup on AWS	Full AWS infrastructure provisioning across four environments using EKS. Includes HA networking, CI/CD pipelines, Kubernetes hardening, container security scanning, FOO observability stack (logging, monitoring, security).
FOO LMS Core Configuration	Baseline deployment and configuration of FOO LMS including organisation structure, offices, BHD currency, chart of accounts, loan product templates for flat-rate and reducing-balance variants, maker-checker permissions, role-based access, and the holiday calendar. Provides the out-of-the-box foundation before Shariah customisation begins.
Murabaha Financing Type & Terminology	Extension of the LMS with a Murabaha-specific data model (cost, profit, selling price), validation rules, and UI/API relabeling of "Interest" to "Profit" across loan product forms, schedules, and transaction responses. Makes FOO LMS a Shariah-compliant system while leveraging its native calculation engine.
Commodity Trading Platform Integration - Capped	Integration layer that captures the commodity contract reference at loan creation, fetches deal information from the external commodity platform, and persists contract ID, product details, purchase/sell dates, and trading status as extended attributes on each financing agreement. Includes a commodity block on the loan details screen.
Core Accounting System Integration - Capped	API-based posting of accounting entries from LMS to BEDE's core accounting system for all portfolio events, disbursement, accrual, collection, settlement, reversal, and write-off, with GL account mapping, error handling and retries, and a reconciliation log between the and the core accounting ledger.
CRM Integration (One-way Customer Read) - Capped	One-way read of customer profile data from the CRM into the LMS portal for display during loan operations and agreement lookup. The CRM remains the master source of customer data, bidirectional synchronisation is explicitly not implemented per client direction.

Deliverables

Deliverable Area	Description Summary
Payment Gateway Integration - Capped	Inbound processing of repayments via payment gateway events and webhooks, including automatic installment allocation, payment confirmation handling (including cheque-clearing status updates passed through from upstream systems), and a reconciliation log between gateway transactions and loan payments.
APIs Exposed for BEDE Backend	REST APIs exposed by the LMS to BEDE's backend services for loan data, schedules, transactions, and restructured-offer retrieval. Includes API authentication, rate limiting, and error code handling. The mobile app consumes these through BEDE's existing backend, there is no direct app-to-LMS contract.
Insurance Handling within Installment (EMI)	Custom scheduling logic that treats insurance as a recurring charge amortised equally across the loan tenure and includes it within the displayed EMI alongside principal and profit. Insurance is managed internally with no external insurer integration, no separate accounting entries, and no dedicated insurance reporting.
VAT & Fee Configuration Engine	Extension of FOO LMS Tax Groups to support VAT-exclusive calculation on all fee types with configurable rates over time, flexible fee definitions (fixed, percentage, one-time, recurring), variable fees with a hard cap (e.g., BD 100), and dedicated fee types for cheque-based surcharges.
Aging, Delinquency & NPA Handling	Configuration of DPD buckets from 1–30 days up to 36 months, automatic cessation of profit accrual at 90+ DPD, inclusion of legal fees and aging penalties in the customer's total outstanding balance, and tracking of the maximum delinquency reached by each customer over the product's lifetime.
Receivable & Deferred Profit Tracking	Dedicated reporting module tracking movement of deferred profit (charged, reversed, new disbursements) and the receivable balance (disbursements, repayments, reversals, write-offs) at the individual agreement level, with real-time visibility of outstanding principal, accrued profit, and unearned profit at any point in time.
Early & Partial Settlement Engine	Automated settlement quote calculation covering outstanding principal, pro-rata accrued profit, and applicable fees. Supports fixed, percentage, and variable-with-cap fee structures plus cheque-based surcharges. Allows customers to pay one or multiple future installments in advance, with auto-closure once all balances are cleared.
Customer Exposure Limit & Multi-Loan Handling	Custom validation hook at the loan origination API enforcing a configurable global exposure limit per customer across all active products in real time. Supports multiple concurrent financing agreements per customer with separate installment schedules for each, aggregated against the exposure cap.

Deliverables

Deliverable Area	Description Summary
CBB & AAOIFI FAS Regulatory Reports - Capped (20 Reports)	Custom reports aligned with Central Bank of Bahrain regulatory templates and AAOIFI FAS-compliant reporting requirements (provisioning, deferred and earned profit recognition). Delivered as standard data exports (CSV/Excel) suitable for manual CBB submission, no direct system-to-system API integration to CBB.
Real-Time Dashboards & Customer Statements	Management and finance dashboards with real-time data refresh (receivables, deferred profit, portfolio yield, delinquency rates, disbursement KPIs), branded PDF customer statement generation on demand, and file-based reconciliation reports against inward/outward bank movements.
Legacy LMS Data Migration & Cutover - Capped	End-to-end migration of up to 10,000 active loan agreements plus required history from the legacy system into FOO LMS. Includes mapping rules, ETL scripts, data cleansing and transformation, two rehearsal cycles, post-migration reconciliation report, and production cutover support with optional parallel run.
UAT Deployment & Support	UAT environment deployment, formal deliverable walkthrough with BEDE, FOO-led System Integration Testing and E2E test execution, and support through two defect-fix cycles during BEDE's user acceptance testing window. BEDE owns test execution and testers mobilisation.
Security Hardening & VA/PT Support	Security hardening of AWS infrastructure (container layer, network, IAM), role-based access control review, and support for one third-party VA/PT audit exercise including remediation of valid findings within the capped effort.
Production Deployment & Go-Live	Orchestrated production deployment with a documented runbook, single go-live cutover window, FOO-led production E2E testing, and handover to post-live hypercare support. All UAT defects must be resolved and signed off before cutover.
Training, Hypercare & Post-Live Support	User guide documentation, remote client training sessions (up to 5 MDs), formal support handover, and a post-live support.

Setup & Foundation

Project Kickoff & Governance Setup

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Kickoff workshop with BEDE stakeholders and PM - Project charter and governance model - Escalation matrix, communication plan - Risk & issue log baseline	- BEDE's internal change-management or procurement workflow - On-site travel or on-premise workshops	- BEDE assigns a dedicated PM and business SME - Decision-makers attend sign-off meetings - FOO tooling is used for PMO

Business Requirements Document (BRD)

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Requirements workshops with BEDE business team - Functional flows covering the full post-disbursement lifecycle - Product configuration parameters (rates, tenures, fees, grace, insurance) - Business rules for aging, penalties, settlement, restructure - Stakeholder-specific report requirements - Final BRD document signed off before SRS begins	- Requirements for Loan Origination - Mobile app customer-facing UX requirements - Wallet, account opening, cash deposit flows - SME financing (future phase) - Re-opening requirements after BRD sign-off (handled via CR)	- BEDE provides complete regulatory references (CBB circulars, FAS standards) - Max 2 iterations before sign-off, further iterations = timeline impact - BRD limited to scope of this RFP (post-disbursement) - Business rules for legal fees & aging penalties to be finalised during BRD

Setup & Foundation

System Requirements Specification (SRS) & Test Cases

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ SRS covering all functional, integration, and non-functional items from BRD ▪ Screen-level specs for the LMS portal (new/modified screens only) ▪ QA test cases reviewed with BEDE ▪ SRS signed off before implementation starts	▪ Detailed design for OOB FOO LMS screens that remain unchanged ▪ Mobile app screen specifications ▪ Design documents for third-party systems ▪ Re-design following mid-implementation scope changes (CR)	▪ BRD is frozen before SRS kickoff ▪ Max 2 iterations on SRS before sign-off ▪ BEDE provides UAT test scenarios, if not, FOO QA provides with 1 iteration ▪ Third-party API specs are available from BEDE before SRS close

Dev / UAT / Prod Environment Setup on AWS

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ AWS infrastructure provisioning: EKS, Aurora (PostgreSQL), S3, Redis, SQS ▪ HA network, container registry, shared storage ▪ Pipelines (CI/CD) and environment promotion flow ▪ Kubernetes hardening and container security scanning ▪ FOO observability stack (logging, monitoring, security)	▪ AWS account purchase, billing, enterprise support contracts ▪ Custom VPC peering with BEDE corporate networks beyond one VPN tunnel ▪ Multi-region active-active deployment (active-passive DR proposed) ▪ Third-party WAF/CDN procurement	▪ BEDE provides AWS account with admin access or pre-provisioned IAM roles ▪ Aurora Global Database for multi-region replication if required (extra cost) ▪ BEDE owns annual infrastructure costs

Setup & Foundation

FOO LMS (Fineract) Core Configuration

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ Base FOO LMS deployment and platform configuration ▪ Organisation, offices, currencies (BHD), CoA setup ▪ Loan product templates: flat rate, reducing balance, grace period variants ▪ Maker-checker permissions & role-based access ▪ Holiday calendar, repayment frequency settings ▪ OOB reports and dashboards baseline	▪ New loan products not defined in BRD (covered via CR) ▪ Multi-currency operations (BHD only for go-live) ▪ Fineract CN (microservices variant) , Fineract 1.x modular monolith is used ▪ Modifications to Fineract core source code (extensions only)	▪ BEDE provides target CoA structure before configuration starts

Shariah & Core Customisation

Murabaha Financing Type & Terminology

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ New financing type on loan product (Murabaha / Tawarruq) ▪ Murabaha-specific data model: cost, profit, selling price ▪ Validation rules aligned with Shariah structure ▪ "Interest → Profit" relabeling on LMS portal UI ▪ Conditional label service / pipe reading financingType ▪ API response enrichment with Murabaha block	▪ Additional Islamic products (Ijarah, Musharakah, Qard Hassan , future phases) ▪ Terminology mapping in downstream systems (mobile app, CRM, core accounting) ▪ Shariah advisory certification or board review ▪ Simultaneous dual-labeling (conventional + Shariah) in same portal session	▪ Murabaha is the only financing type at go-live ▪ Other financing types are configurable extensions for future phases ▪ Downstream systems map terminology on their side ▪ BEDE provides Arabic/English label glossary

Shariah & Core Customisation

Commodity Trading Platform Integration

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Commodity reference input at loan creation (contract ID) - Fetch & auto-populate deal info from commodity platform API - LMS datatable extension to persist contract ID, product, dates, status - Display commodity block on loan details screen - Association between each financing agreement and its commodity transaction	- Development or modification of the commodity trading platform - Real-time bidirectional sync (one-way read at creation + status updates) - Commodity fee calculation engine beyond capture of reported fees - Replacement of commodity platform	- Commodity API specs, auth, sample payloads provided by BEDE before build - Commodity API is stable, retry/failure handled gracefully with manual intervention - Sync frequency and reconciliation rules defined in SRS - Integration effort capped, additional effort requires CR

Core Accounting System Integration

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- API-based posting of accounting entries to the core accounting system - Events: disbursement, accrual, collection, settlement, reversal, write-off - Error handling, retries, reconciliation log	- Modification of the core accounting system itself - Multi-ledger / multi-entity posting - Historical GL data backfill beyond migration scope - Real-time GL balance queries from LMS to core	Core accounting APIs documented and available in stable non-prod environment

Shariah & Core Customisation

CRM Integration (One-way Customer Read)

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- One-way read of customer profile data from CRM into LMS portal - Customer search and basic demographic display on LMS UI - CRM as master source of customer data	- Bidirectional sync - Customer onboarding workflows - KYC/AML processes and document management - Modifications to the CRM system	- CRM API specs, auth, sample payloads provided by BEDE - CRM APIs are stable in a non-prod test environment

Payment Gateway Integration

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Inbound repayment processing from payment gateway - Event/webhook handler to mark installments as paid - Reconciliation log between gateway transactions and loan payments	- Gateway merchant onboarding or procurement - Cheque validation/clearing (handled upstream) - Payment initiation by customer from LMS (customer payments happen outside LMS) - Card tokenisation / PCI-DSS scope on LMS side (not storing card data)	- Gateway APIs documented and stable - Cheque clearing handled by BEDE upstream, LMS receives confirmed payment only

Shariah & Core Customisation

APIs Exposed for BEDE Backend

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ LMS REST APIs for backend services to fetch loan, schedule, transaction data ▪ Expose restructured offers for customer review through existing channels ▪ API authentication, rate limiting, error codes	▪ Direct mobile app integration (mobile connects via BEDE backend) ▪ Mobile app UI/UX development ▪ Biometric, OTP, e-signature workflows on the app ▪ Customer acceptance flow UX for offers ▪ Wallet integration	▪ Mobile app backend handles customer acceptance and passes the signed offer to LMS ▪ BEDE backend team consumes LMS APIs, no direct app-to-LMS contract

Financial Modules & Compliance

Insurance Handling within Installment (EMI)

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Insurance as a recurring charge, amortised equally across tenure - Insurance percentage configurable at product and loan level - Insurance included within the EMI display (Principal + Profit + Insurance) - Insurance treated as part of the installment	- External insurer integration - Separate insurance-specific accounting entries or dedicated reports - Refund handling on early settlement - Insurance policy document management	- Insurance is managed internally by BEDE, not passed to a third-party insurer - VAT treatment on insurance aligned with general fee rules - Custom scheduling logic or advanced payment allocation will be used

VAT & Fee Configuration Engine

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Flexible fee definitions (fixed, percentage, recurring, one-time) - VAT-exclusive calculation applied on top of base fee - Configurable VAT rates over time per fee type - Automatic VAT posting and fee application - Variable fees with a configurable hard cap	- Country-specific tax registration or filing automation - Complex multi-tier tax stacking (only single VAT per fee) - Historical VAT adjustment on existing loans	- VAT rate changes are applied prospectively unless BRD states otherwise - Fee types and caps finalised in BRD, CR applies to further types - FOO LMS Tax Groups used, extended with capped-fee custom rule

Financial Modules & Compliance

Aging, Delinquency & NPA Handling

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Configurable DPD buckets from 1–30 days up to 36 months - Auto-cease of profit accrual - Legal fees & aging penalties included in customer total outstanding - Early-settlement penalty tracking - Historical tracking of max delinquency reached over product life - Delinquency dashboards	- External collections agency integration - Legal case management system - Automated dunning letters or customer communications - ECL/IFRS9 probabilistic modeling (static provisioning only, full ECL is a future phase)	- Formulas for legal fees & aging penalties finalised jointly during BRD - Bucket thresholds configured during setup, not via UI at runtime - ECL support is data-layer only, no probabilistic engine

Receivable & Deferred Profit Tracking

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Deferred Profit movement report: profit charged, closed-account reversals, new disbursements - Receivable balance movement: disbursements, repayments, reversals, write-offs - Real-time outstanding principal, accrued profit, unearned profit tracking - Agreement-level drill-down - Dashboards reflecting receivable & deferred profit	- Predictive analytics or forecasting of receivables - Variance analysis against budget - Cross-subsidiary consolidation reporting	- LMS upfront-accrual accounting is used for Murabaha - Reports built via Metabase - Up to 10 reports configured, additional reports via CR bucket

Financial Modules & Compliance

Early & Partial Settlement Engine

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Automated early settlement quote: outstanding principal + accrued profit + applicable fees - Configurable early-settlement fee: fixed, percentage, or variable-with-cap - Cheque-based settlement surcharge - Partial settlement: pay one or multiple future installments in advance - Auto-closure on full settlement	- Customer-facing settlement quote UI on the mobile app - Negotiated waiver workflows	- Pro-rata profit accrual to settlement date - Fee caps and cheque surcharge amounts configured in BRD

Customer Exposure Limit & Multi-Loan Handling

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Support multiple active financing agreements per customer - Configurable global exposure limit per customer - Real-time validation at loan creation / approval - Sum across all active products not to exceed the cap	- Credit scoring model - Bureau (BCRB) integration for external exposure - Dynamic limit adjustment based on behaviour	- Exposure limit is a static configured value per customer / risk tier - Custom validation hook implemented on loan-origination API

Financial Modules & Compliance

CBB & AAOIFI FAS Regulatory Reports - Capped to 20 Reports

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ Custom reports aligned with CBB templates ▪ AAOIFI FAS-compliant reports (provisioning, deferred/earned profit) ▪ Standard data exports (CSV/Excel) suitable for manual submission to CBB ▪ Capped delivery of an agreed number of regulatory reports	▪ Direct system-to-system API submission to CBB ▪ Regulatory certification or sign-off (client responsibility) ▪ Reports not defined during the SRS phase (handled via CR) ▪ ECL / IFRS9 probabilistic engine	▪ BEDE provides final CBB templates and FAS formats during implementation ▪ Number of reports agreed in BRD and frozen before build ▪ The scope includes up to 20 reports. Any additional reports will be considered out of scope and charged separately based on the applicable man-day rate.

Real-Time Dashboards & Customer Statements

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ Management & finance dashboards with real-time data refresh ▪ Receivables, deferred profit, portfolio yield, delinquency, disbursement KPIs ▪ Branded customer statement (PDF) generation ▪ Bank-movement reconciliation report (file-based) ▪ Export to CSV/Excel	▪ Third-party BI tool licensing (Tableau/PowerBI) ▪ Customer self-service statement UI on the mobile app ▪ Proactive statement email/SMS distribution	▪ Real-time = sub-minute refresh acceptable (not strict streaming) ▪ Statement template provided by BEDE in BRD

Migration & Delivery

Legacy LMS Data Migration & Cutover

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ Migration methodology document and mapping rules ▪ ETL scripts to migrate up to 10,000 active loan agreements and associated history ▪ Data cleansing and transformation to Fineract schema ▪ Reconciliation report post-migration ▪ Mock migration cycles before go-live ▪ Cutover support during production migration	▪ Cleanup of data quality issues in source systems ▪ Migration of closed/inactive loans beyond agreed historical retention ▪ Data archival strategy for the legacy LMS ▪ Parallel-run operations beyond 4 weeks (additional via CR) ▪ Data model reverse-engineering of legacy system	▪ BEDE provides the legacy schema, field mapping, and sample extracts ▪ Data available in a single consolidated format or via extract pipeline ▪ Source data is in a reasonable quality state, data cleansing effort is minimal ▪ Parallel-run of up to 4 weeks at BEDE discretion, 2 reconciliation passes ▪ Capped Efforts

UAT Deployment & Support

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ Deployment to UAT environment ▪ Deliverable walkthrough with BEDE ▪ FOO-led System Integration Testing (SIT) ▪ E2E test execution by FOO QA ▪ Defect fix cycles during UAT (2 cycles)	▪ UAT test execution by BEDE team ▪ User acceptance criteria definition beyond what SRS covers ▪ Support for ad-hoc UAT environment changes outside defect fixes ▪ Defects originating from third-party systems	▪ BEDE provides UAT test cases, else FOO QA provides (1 review iteration) ▪ BEDE manages UAT testers and execution ▪ UAT completes within agreed window, extensions impact timeline/budget

Migration & Delivery

Security Hardening & VA/PT Support

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Security hardening of AWS infrastructure (container, network, IAM) - Support for one VA/PT exercise including remediation of valid findings - Role-based access control review - Compliance with data security standards as defined in the RFP	- The VA/PT audit engagement itself (to be engaged by BEDE) - PCI-DSS certification (no card data stored in LMS) - Re-testing for invalid/low-quality findings beyond the capped effort - SOC 2 or ISO 27001 certification	- VA/PT auditor delivers actionable, valid findings - Poor-quality audits consume extra budget, CR applies - One full VA/PT cycle is within the 5-MD cap

Production Deployment & Go-Live

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
- Production deployment with documented runbook - Go-live cutover orchestration (single window) - Hypercare hand-off to post-live support	- Multiple go-live windows / phased rollouts not agreed in BRD - Customer communications and marketing 24x7 deployment support beyond the cutover window	- Single go-live event - All UAT defects resolved and signed off before cutover

Migration & Delivery

Training, Hypercare & Post-Live Support

IN SCOPE	OUT OF SCOPE	ASSUMPTIONS
▪ User guide documentation ▪ Client training sessions (up to 5 MDs) ▪ Support handover session ▪ Post-live support	▪ Ongoing training beyond initial rollout ▪ Knowledge transfer to a third-party support vendor ▪ New features or change requests (separate CRs) ▪ NOC/SOC services unless optional annex purchased ▪ Hardware, travel, and out-of-pocket expenses	▪ Training is remote by default, on-site is extra ▪ M&S renews annually per separate SLA

Infrastructure

Environments

FOO will use BEDE's existing infrastructure.



Client and 3rd Party Integrations

FOO will need to integrate with

- CRM Provider
- Core Accounting System
- Payment Gateway
- Commodity trading platform

Due to the lack of documentation, integration efforts for each provider are capped. **Any additional efforts** required will be assessed once all the required documentation is available.



Client and 3rd Party Integrations

Third party APIs should be ready, documented and tested.

For each integration **FOO** will be setting up the full API testing needed as a first step, simulating the flow of API integrations, and including any dummy data as needed to have a full test of the integration service.

All issues that arise during this process will be raised to the client and no additional effort should be done by **FOO** until all issues are resolved.

Once the above step is completed the integration will be developed on the application level.

Any issues or changes that arise after the first sign off on the integration APIs will be estimated and charged separately (in case any effort is needed from FOO's side for debugging or changes in flow...)

SMS & Email gateways (SMTP) to be provided by the client.

The below is required to be provided for integration ahead of time

- List of expected APIs
- Documentation of the APIs
- Request/Response samples
- Access to the APIs through one of the below methods:
 - Public access
 - IP Whitelisting (FOO IP)
 - VPN site to site tunnel
- Fixed data needed for APIs (IDs, passwords, tokens, keys, etc..)
- Test data available for all APIs to be able to perform a full flow
- Error codes returned in APIs to be handled
- APIs to be provided on a stable environment
- Regular updates on any changes applied on the APIs



Assumptions

Assumptions and Constraints

Based on the solution requirements for this project, the following assumptions and constraints are included

- LMS Migration is capped as it needs a separate discussion and further clarifications.
- Integration with BEDE's current wallet is out of scope for this RFP and will be treated as separate project.
- Loan offer creation, internal approval hierarchies, and customer acceptance mechanisms (biometric, OTP, e-signature) on the mobile app are outside the LMS scope.
- Account opening, digital wallet management, and cash deposit acceptance are out of scope.
- Flat-rate is the primary profit calculation method, reducing-balance is supported for disclosure, comparison, and future product configurations.
- Profit rates are configurable at both product and individual customer level.
- SME financing is a future-phase requirement and will be addressed separately.
- Grace periods (principal-only, interest-only, and full) are configurable per product.
- Multiple active financing agreements per customer are supported with separate installment schedules.
- Insurance is treated as part of the installment structure (EMI) and managed internally within the LMS, no external insurer integration is required.



Assumptions and Constraints

- No separate insurance-specific accounting entries or dedicated insurance reporting are required.
- All fees are treated as VAT-exclusive, VAT is calculated and applied on top of the base fee amount.
- VAT rates and VAT applicability are configurable over time at the fee level.
- There is no predefined limit on the number of fees that can be configured, both one-time and recurring fees are supported.
- Partial settlement and advance installment payments are supported.
- Loan closure is system-driven and occurs automatically once all outstanding balances are cleared. Specific business rules, triggers, and formulas for legal fees and aging-based penalties will be defined jointly with BEDE during implementation.
- Any adjustment to the OOB Rules impose extra fees.
- Specific CBB report templates, data structures, and submission frequencies will be provided by BEDE during implementation, direct system-to-system submission to CBB is not required, standard data exports are sufficient.



Project Deliverables

Capped Efforts

Throughout this document, multiple activities are labeled as **capped to X MD** (man days).

This indication highlights the importance of a predetermined limit on the amount of time and efforts that will be spent on a particular task. This concept is often used for activities whose costs cannot be accurately estimated and can vastly vary from an implementation to another.

The key aspects of capped estimates lie in:

- **Fixed Budget:** There is a capped / maximum budget allocated for the specified effort, beyond which the client will be charged.
- **Defined Scope:** The tasks under capped efforts are clearly defined, ensuring that any amendments on the scope will require additional charges.
- **Transparency:** While some activities can be clearly defined and estimated, others remain high level up until the implementation phase where some business and technical decisions get clearer, the reason why FOO remains transparent vis-a-vis the client by capping such activities.



Project Deliverables

For instance, capped efforts are usually mentioned for activities such as:

- **Requirements Gathering & Scoping Activities:** The requirements gathering phase might vary a lot in terms of timeline and effort based on the parties involved in this exercise, the readiness and maturity of the requisites.
- **Integration Activities:** The implementation of integration layers between FOO Backend Components and other 3rd party systems can only be high level estimated since all detailed technical documents for the 3rd party APIs are missing during the writing of this document. Many variables can highly increase these tasks such as: 3rd party APIs accessibility, connectivity, discovery, and testing.
- **UI/UX Activities:** User journeys and design, when applicable, are highly subject to change based on multiple Q&A sessions with the client. Hence, such activities are limited in terms of iterations and capped in terms of efforts.



Team and Organization

A Project Manager from the client side is expected to be assigned to the project, to help resolve any issue arising.

The client is expected to include key decision makers for the design phase and the sign off on all deliverables.

It is assumed the **FOO tools** will be used for issue and risk tracking and communication (Jira, Google Meet, Slack etc. or similar)



QA And Testing

The client is responsible for the creation of the User Acceptance Test scenarios and test cases and their execution.

The client is expected to furnish **FOO** with all necessary assets to enable and expedite the testing process. These assets may include, but are not limited to, national IDs, passports, cards, and any relevant bill payment identifiers.

The client should deliver the UAT test cases to FOO, to be used by the **FOO QA department** for internal validation prior to delivery for testing. If UAT test scenarios are not provided, FOO QA team will share the test cases with the client and will be limited to 1 iteration with the client to finalize the list of test cases.

FOO is responsible for the creation of the System Integration Test (SIT) scenarios and test cases and their execution.

The client is expected to manage the UAT cycle, including client testers mobilization, tracking the test execution, raising issues detected to FOO and coordinating the fixes, merge and retest.



Out of Scope

Below is a list of what is out of scope including features mentioned in the Product deck:

- Any integrations with third parties not mentioned in this proposal
- Custom development services that are not mentioned in this proposal
- On-site hardware or software installation
- Out-of-pocket and reasonable expenses, including hardware and related supplies, travel expenses, etc..



Development Plan

Methodology

The development methodology followed for this project is **agile methodology**.

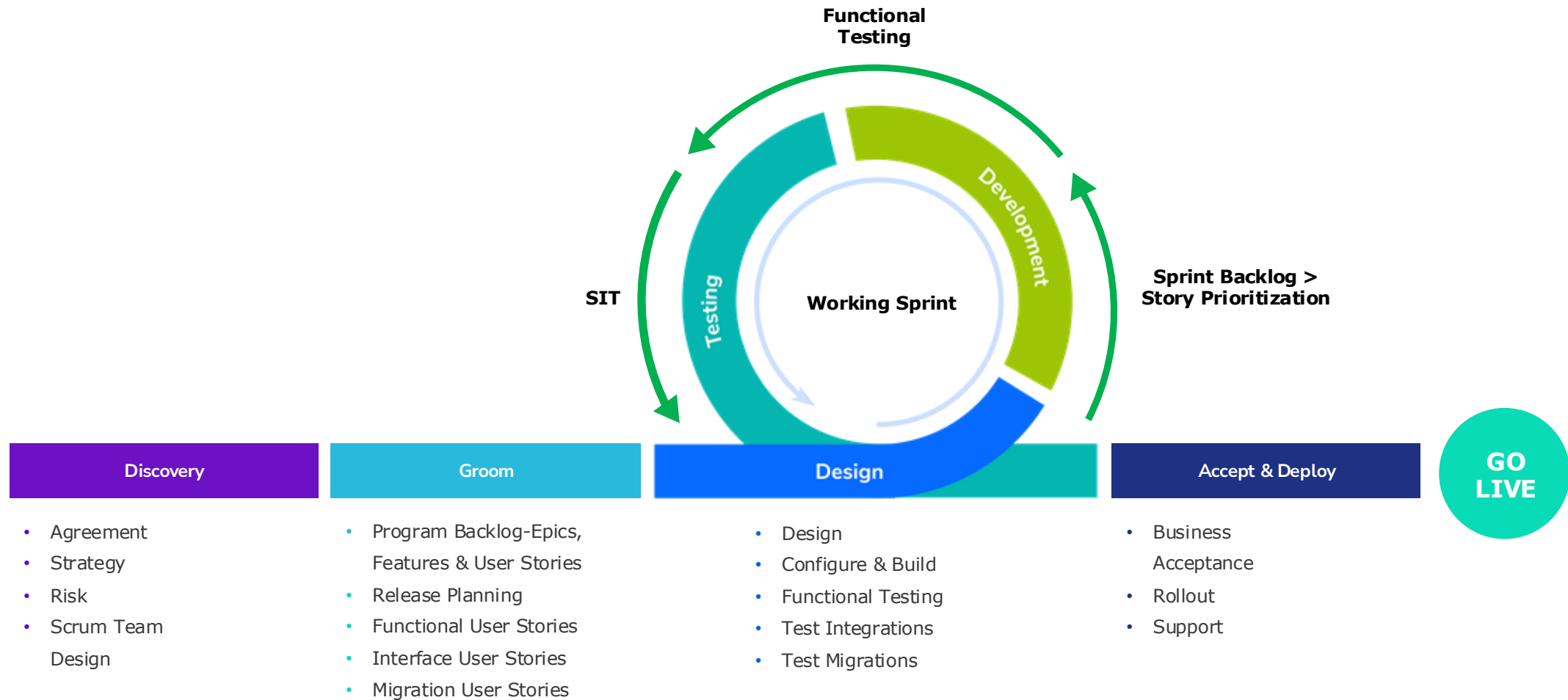
Will be split into sprints that will cover 1 month and will contain the features planned.

The features in each sprint can be rearranged based on the client priorities.

All features included regardless of their position in the sprint are all based on a single release which means one production deployment at the end.



Methodology



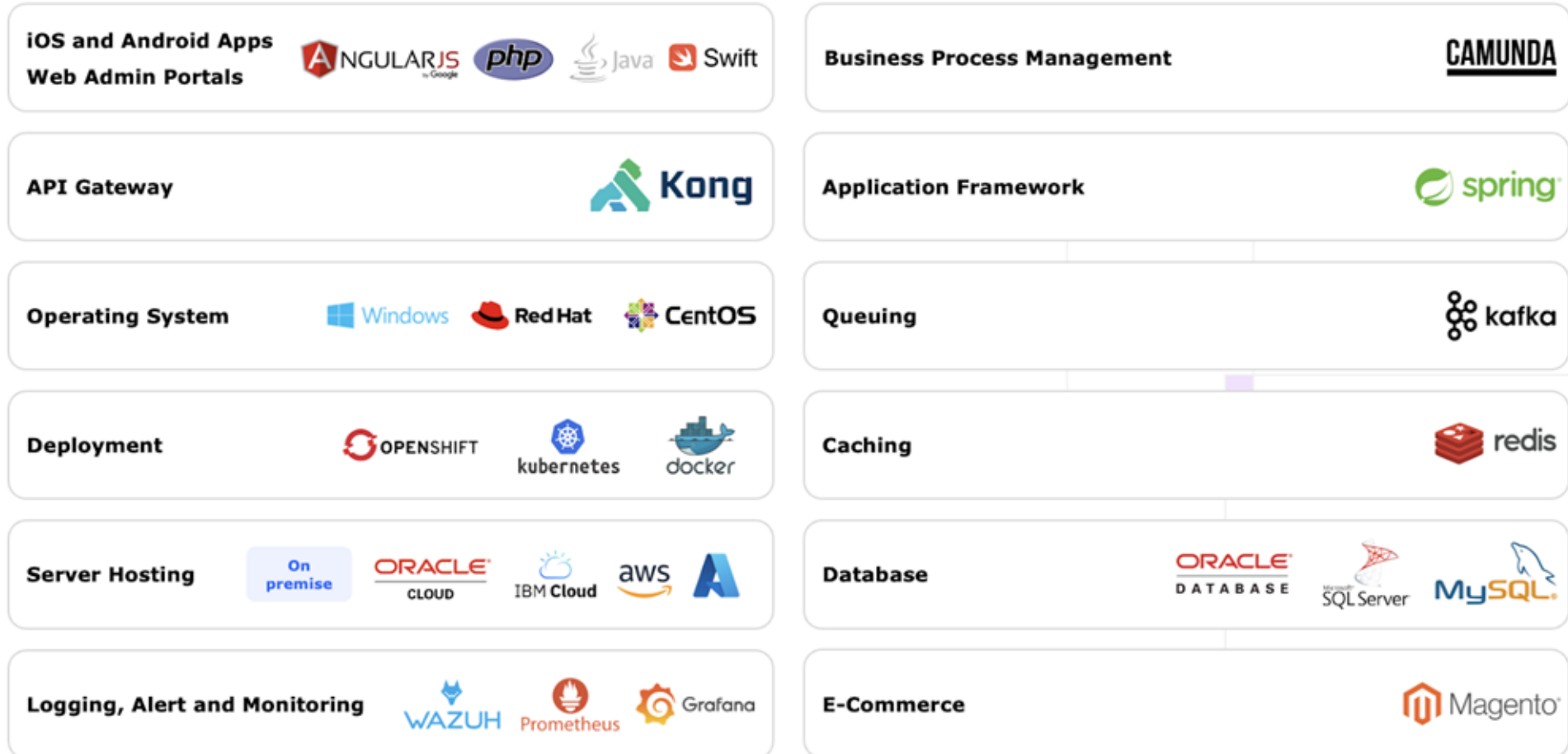
Team Members

The following will be the team involved in the development of the solution in each sprint:

- PM
- PTL
- BA
- Java Developer
- Java Integrator
- PHP Backend Developer
- Web Portal Developer
- QA
- Sysadmin / Devops



Technology Stack



Project Plan

The project duration is estimated to be

4 months

given the platforms in scope and the related assumptions.

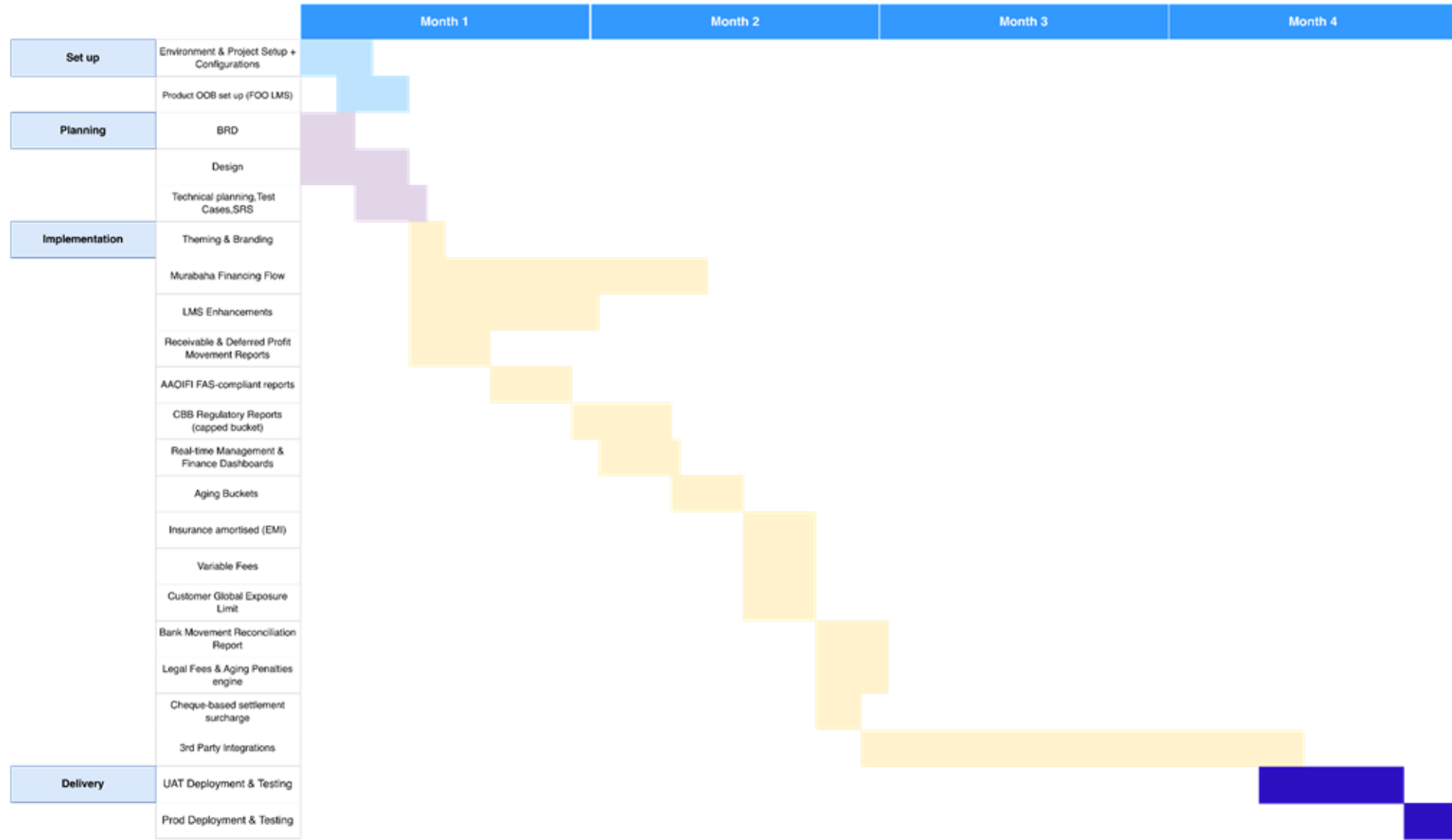
The below image is the high-level project timeline.

A detailed Project Implementation Plan will be shared with the client upon completion of the requirements gathering.

PM and PTL efforts are charged based on the duration of the project, if the duration of the project is extended additional will be incurred based on the manday rate shared in the commercial proposal.



High Level Timeline



Project Milestones

Each sprint release will be a milestone, for each sprint we will need the requirements and planning to be completed and signed off before the start of the next sprint.



Project Timeline and Budget Impact Considerations

- If project timelines are extended beyond the initially agreed schedule, additional budget will be required to cover the extended period. This includes costs related to resources, administrative overhead, and potential re-allocation expenses.
- If the SRS phase requires more time and iterations than initially planned, the project timeline will be affected. If no additional changes are made to the scope, the timeline will be extended by the same period needed to complete the SRS. Any new changes will result in a re-assessment to evaluate their impact on the final timeline and allocated or capped budget. Any new changes requested during implementation will also be assessed for potential impact on timeline and budget.
- If the UI/UX phase requires more time and iterations than initially planned, the project timeline will be affected. If no additional changes are made to the scope, the timeline will be extended by the same period needed to complete the UI/UX. Any new changes will result in a re-assessment to evaluate their impact on the final timeline and allocated or capped budget. Any changes requested during implementation will be assessed for potential impact on timeline and budget.

Project Timeline and Budget Impact Considerations

- If the infrastructure is provided by a third party, any delays in delivering any of the agreed environments will impact the timeline. Additionally, any issues with the connection between FOO servers and third-party integration endpoints will affect the timeline and potentially the allocated or capped budget if FOO is responsible for the connection testing.
- If third-party APIs are not provided on time or do not function as expected, additional budget will be required to address delays in integration, troubleshooting, testing, and potential rework. This includes any additional development, review, and testing efforts. Timelines will also be impacted based on the duration of the delay and resource reallocation.
- If the UAT phase takes longer and requires more iterations, a re-assessment of resource availability will be done, and the allocated or capped budget will be adjusted to cover extended testing efforts, additional iterations, feedback incorporation, and retesting.



Project Timeline and Budget Impact Considerations

- Poor or low-quality third-party VA & PT (Vulnerability Assessment and Penetration Testing) auditing will impact the initial time and allocated or capped budget to re-check and evaluate any invalid raised issues. If the allocated or capped budget is affected by the additional efforts mentioned above, the client has the option to purchase a manday bucket, which will be calculated based on the allocated manday rate. Available buckets are 5 MD, 10 MD, or 15 MD.
- Any new regulatory compliance requirements not listed in the initial commercial or technical proposal will be subject to timeline and budget assessment.
- In case of a project stoppage or blockage due to external factors beyond FOO's control, an evaluation will be conducted to determine the necessary timeline adjustment and budget impact. Mainly this is due to potential resource reallocation.
- If a change request (CR) remains in User Acceptance Testing (UAT) for an extended period without being promoted to production, bringing it back to life will require additional time and effort. This includes reallocating team members, onboarding as needed, retesting the solution, and addressing any potential fixes due to changes that have occurred in the meantime. These efforts will impact the overall budget and require additional funds.





Support

Maintenance and Support

Providing the basis for managing relationships between **FOO** **and the Client**, service level management deals with how user service requirements are understood and handled.

Many benefits can be realized from a well-implemented service level management discipline.

Check **Annex 2** for full details in **SLA** document



Upgrades

Upgrades to product versions, technology versions, security updates will be included as part of the M&S budget, and all backend upgrades will be handled by **FOO**.



Logging

We use a **centralized log management** tool built to open standards.

The tool's connectivity and interoperability seamlessly collect, enhances, stores, and analyzes log data.

Coupled with a logs analysis tool (for gathering the logs from multiple servers), the log management tool can parse the logs in real time and send alerts if a certain predefined pattern occurs.

The log management tool is to be installed on an independent server and an instance of the log analysis tool to be installed on all the servers that need to be monitored.



Logging

Use cases

- **Code Exceptions**

The tool can read application logs.

We can set a specific pattern so that we would be alerted by email in case exceptions occur.

for example, if a certain exception appears in the logs 5 times in 1 minute this could mean that there is something about to go wrong.

It can also be configured to send an alert for certain types of exceptions.

- **System logs**

The tool can log and monitor all system logs to identify the following:

- Successful ssh login attempts
- Failed ssh login attempts
- sudo access or root access

- **File Monitoring**

The tool can monitor the files to detect and alert on any change on monitored files like server or application configurations.

- **Log querying**

The tool has a GUI used to query the logs and it manages indices to be able to query base on certain parameters (e.g. user, device, request id, transaction id.)



Monitoring

We use a centralized monitoring system where clients of that tool are installed on each server to monitor the server availability and usage.

It can be configured to send alerts on certain scenarios or thresholds.

Monitoring Use cases

- CPU usage
- Memory
- Disk space
- Service availability (like Apache, MySQL, Wildfly, SSH,...)
- Connections (connection between servers or to certain APIs or modules)



Change Management

Changes that are not part of the requirements discussed, or that are not covered in the requirement documents will be managed in a separate change request document (**CR**).

The **CR document** will be considered binding once signed by both parties and will cover the cost of implementation, as well as the date & time for delivery.

CRs will have a validity date after completion if the client decides not to go live right away with the CR.

After the expiration date, the CR will become dormant and will require efforts to be assessed to reactivate and merge with existing features if applicable.



Acceptance Criteria

Client will accept each Project Deliverable upon its completion and will provide the company with a corresponding signed certificate (**the "Acceptance Certificate"**) – or within **5 days** notify the Company of any reasonable objections against Acceptance of the relevant Project Deliverable.

Company will advise an appropriate rectification period within 2 days of receiving such a notification and will re-submit within the rectification period as advised. The overall cycle can be repeated **up to 2 times**.

Failing to raise objections and after the **5 days period**, the Project Deliverable will be considered as accepted.

Any third party task preventing testing and acceptance will impact the budget and timeline of the project. Payments related to deliverables that could not be tested due to a third party intervention will be released after **5 business days** from the delivery.

The testing and acceptance can be resumed once the third party issue is resolved.



Annexes

Annex	Document Title	Document Name
Annex 1	Company Profile	FOO Company Profile v1.14.pdf
Annex 2	Service Level Agreement	FOO - Service Level Agreement v2.13.pdf

Let's Get Started!